

8ai	Heat lost by water = heat gained by mercury $m_{\text{water}}C_{\text{water}}\Delta T_{\text{water}} = m_{\text{mercury}}C_{\text{mercury}}\Delta T_{\text{mercury}}$ $(18.7)(4.18)(37.4 - T) = (6.94)(0.140)(T - 23.0)$ $T = 37.2\text{ }^{\circ}\text{C}$	A	C1 C1 A1
8aii	The glass of the thermometer OR the beaker containing the water have negligible heat capacity or gained negligible heat.	A	B1
8aiii	Use a liquid with a lower (specific) heat capacity (than mercury), or Use a smaller mass of mercury.	A	B1
8aiv	It depends on properties of a real substance, and 0 °C is not absolute zero	D	B1 B1
8av	Ideal gas	E	B1
8bi	Internal energy is the sum of the kinetic energy due to the random motion of the molecules and the potential energy E_p due to intermolecular forces between them. For ideal gas, no forces between molecules, so the potential energy of molecules is zero. Hence internal energy of the gas is equal to the total kinetic energy of the molecules	A	B1 B1 A0
8bii	Increase in internal energy = Q + work done, Constant volume so no work is done.	A	B1 B1
8biii	$Q = Nmc\Delta T$ $U = (3/2) NkT$, so $\Delta U = (3/2) Nk\Delta T$ $Q = \Delta U$, so $Nmc\Delta T = (3/2) Nk\Delta T$ Hence, $c = \frac{3k}{2m}$	D	M1 M1 A1 A0
8biv	Gas expands and does work (against the atmosphere pressure) (For the same temperature rise), more (thermal) energy is needed, so larger specific heat capacity.	A	M1 A1
8bv	Speed of air molecules decreases on impact with moving piston. Since the kinetic energy (of molecules) is proportional to (thermodynamic) temperature, Kinetic energy (of molecules) decreases, so temperature decreases.	D	B1 B1 B1

9a	The Magnetic Flux Linkage in a coil is the product of the area of the coil and the	F	M1
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